

30th Jan 2024 | CS6370

$$\text{Norm TF} = \frac{\text{TF}(t, d)}{\text{Max TF in the doc}}$$

- TF is Normalized TF
↓
no. of times t appears in d → $\frac{\text{count of } t \text{ in } d}{\text{total terms in } d}$

- cosine similarity

↓
compute all zoo products

- but, why vector spaces? (are being used in the '0' → linear alg.)

- very frequent terms tend to receive a low wt. - a of 0 in this case.

- Model correctly predicts that very common terms occurring in many documents in a collection are not good discriminators of relevancy. This reasoning is based on global info (IDF term). That is why this is better than term count model.

- Vectors → reduces lots of computaⁿ (resource intensive)

- Good read → Great Ideas in Computer

Science

(Verb + Noun + Rule)

(5) ←

how to formalize what is lang.?

.NLP

↙ ↘

NLU

NLG

(understand) (generaⁿ)

how humans understand language is even more harder?

- Distributed Computational Semantics

regarding evaluations $\rightarrow$ efficiency
 $\rightarrow$ basic things $\rightarrow$ effectiveness

Imp. found. measures

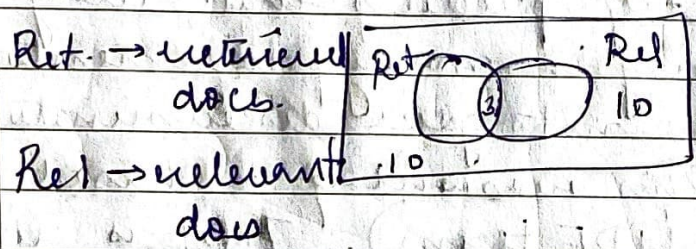
Precision $\rightarrow$ of all items predicted as true, how many are true posits

$$\frac{TP}{TP + FP}$$

Recall $\rightarrow$ of all actual positives, how many positive model could find

$$\frac{TP}{TP + FN}$$

eg. we get some query Q



$$\text{Precision} = \frac{|Ret \cap Rel|}{|Ret|}$$

$$\text{Recall} = \frac{|Ret \cap Rel|}{|Rel|}$$

3000 docs $\rightarrow$ 10 relevant
 $\rightarrow$ 10 retrieved

things also are relevant based on which application we are trying for.

why harmonic? Recall-centric $\rightarrow$ what is likely to be sys. suspect, just catch him.

clear balance

$$F1 \text{ measures} \rightarrow \text{harmonic mean of precision \& recall.}$$

$$\left(\frac{2PR}{P+R} \right) = \left(\frac{2}{\frac{1}{P} + \frac{1}{R}} \right)$$

weighted F1 $\rightarrow$ compute F1 for each class, measure then avg. them weighted by the no. of true samples (support) of each class

imbalanced datasets

$$wt. F1 = \frac{\sum F1 \times S_i}{\sum S_i}$$

- what are things that might get missed by precision & recall in IR process?

→ degree of relevance

→ this is only for a single query, for similar of such queries, avg. out.

→ retrieved documents are not yet ranked (which is crucial) → highly ranked should be at the top.